

Scenario 1: Delivery Time Analysis for an E-commerce Company

An e-commerce company tracks delivery times (in minutes) for **15 orders**:
[25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95]

The company wants to analyze the delivery performance using percentiles and detect if there are any unusual delivery times.

✔ Question 1: Calculate Q1 and Q3

- **Q1 (25th Percentile)** → 25% of the data is below this value
- **Q3 (75th Percentile)** → 75% of the data is below this value

□ **Formula:**

$$Q1 = 0.25 \times (n+1) \quad Q1 = 0.25 \times (n+1) \quad Q3 = 0.75 \times (n+1) \quad Q3 = 0.75 \times (n+1)$$

- **Q1** = 4th value → **40**
- **Q3** = 11th value → **75**

✔ Question 2: Find the IQR

- **Interquartile Range (IQR)** measures the spread of the middle 50% of data.

□ **Formula:**

$$IQR = Q3 - Q1 = 75 - 40 = 35 \quad IQR = Q3 - Q1 = 75 - 40 = 35 \quad IQR = Q3 - Q1 = 75 - 40 = 35$$

✔ Question 3: Detect Outliers

- **Outliers** are extreme values.
- Use these formulas to find boundaries:

□ **Formulas:**

$$\text{Lower Bound} = Q1 - 1.5 \times IQR \quad \text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR \quad \text{Upper Bound} = Q3 + 1.5 \times IQR$$

$$\text{Bound} = Q3 + 1.5 \times IQR$$

- **Lower Bound** = $40 - 1.5 \times 35 = -12.5$ $40 - 1.5 \times 35 = -12.5 \rightarrow$ No lower outliers
- **Upper Bound** = $75 + 1.5 \times 35 = 127.5$ $75 + 1.5 \times 35 = 127.5 \rightarrow$ No upper outliers. ✓ **Conclusion:** No outliers in this dataset.

Scenario 2:

A teacher is analyzing the mathematics scores of students in her class. The scores are: [45, 50, 55, 60, 60, 62, 63, 65, 90, 95]

Question:

- Calculate the mean, median, and mode of the scores.
- Explain why the median might be a better representation than the mean in this case.

Answer:

- **Mean:** $\frac{45+50+55+60+60+62+63+65+90+95}{10} = 65.5$
- **Median:** Middle value = $\frac{60+62}{2} = 61$
- **Mode:** 60 (Occurs twice)

Explanation:

- The **median** is a better measure here because the high outliers (90 and 95) skew the mean. The median is not affected by extreme values, providing a better reflection of student performance.

Scenario 3:

A grocery store manager tracks how many customers visit the store daily for a month:
[5, 10, 8, 15, 20, 5, 12, 14, 10, 18]

Question 1:

- Create a frequency distribution table for this data.

Answer:

Number of Customers	Frequency
5	2
10	2
8	1
12	1
14	1
15	1
18	1
20	1

Scenario 4:

A real estate model has three variables:

- House Size
- Number of Rooms
- Number of Bathrooms

Question 1:

- How can you detect multicollinearity?

Answer:

- **Logic:**
 - Calculate the Variance Inflation Factor (VIF).
 - $VIF > 10$ indicates multicollinearity.
- **Answer:** High VIF means the variables are correlated, impacting model accuracy.

Scenario 5:

A company made a new medicine to lower blood pressure. They gave it to one group and gave a fake pill (placebo) to another group.

Question 1:

How can the company check if the new medicine works?

Solution:

- **Make a guess (Hypothesis):**
 - **H0:** The medicine doesn't lower blood pressure.
 - **H1:** The medicine lowers blood pressure.
- **Do a T-Test:**
 - Find the **p-value** (a number that shows how likely the result happened by chance).
 - If **p-value** < 0.05 , it means the medicine likely works.

- **Final Answer:**
 - If the **p-value** is small, the medicine is effective.



Scenario 6: Identifying Outliers in Sales Data

A company wants to find any unusual spikes in sales.

Question 1:

How can the company detect outliers in their sales data?

Logic:

- Step 1: Calculate the Interquartile Range (IQR).
- Step 2: Identify outliers using the formula:

$$\text{Outliers} = (\text{Data} < Q1 - 1.5 \times \text{IQR}) \text{ or } (\text{Data} > Q3 + 1.5 \times \text{IQR})$$

Scenario 7: Understanding Customer Satisfaction

A restaurant conducted a survey to rate customer satisfaction on a scale of 1 to 5:

[5, 4, 4, 5, 3, 4, 5, 2, 4, 3]

✓ **Question 1:** How can the restaurant summarize the overall satisfaction?

Answer:

- Find the Mode to see the most common rating.
- Calculate the Mean and Median for further insights.
- If most ratings are 4 or 5, satisfaction is generally high.

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